2.33 Integration: Reversing the Chain Rule Part 1

Worked Solutions 1–21

1 a)

$\int (x+1)^3\,dx = \frac{1}{4}(x+1)^4 + c$

1 b)

$\int (x-4)^6\,dx = \frac{1}{7}(x-4)^7 + c$

1 c)

$\int 6(2+x)^7\,dx = 6 \times \frac{1}{8}(2+x)^8 + c = \frac{3}{4}(2+x)^8 + c$

1 d)

$$\int (10(x-1)^4 - 4(x-3)^5)\,dx = 10 \times \tfrac{1}{5}(x-1)^5 - 4 \times \tfrac{1}{6}(x-3)^6 + c = 2(x-1)^5 - \tfrac{2}{3}(x-3)^6 + c$$

2 a)

$\frac{dy}{dx} = (x-3)^{3/4} \Rightarrow y = \frac{4}{7}(x-3)^{7/4} + c$

2 b)

$\frac{dy}{dx} = (9+x)^{-3} \Rightarrow y = -\frac{1}{2}(9+x)^{-2} + c$

2 c)

$\frac{dy}{dx} = \sqrt{x+7} \Rightarrow \frac{dy}{dx} = (x+7)^{1/2} \Rightarrow y = \frac{2}{3}(x+7)^{3/2} + c$

2 d)

$\frac{dy}{dx} = \frac{4}{\sqrt{5+x}} \Rightarrow \frac{dy}{dx} = 4(5+x)^{-1/2} \Rightarrow y = 4 \times 2(5+x)^{1/2} + c \Rightarrow y = 8(5+x)^{1/2} + c$

3)

$$\frac{dy}{dx} = \frac{6}{(x-2)^5} \Rightarrow \frac{dy}{dx} = 6(x-2)^{-5} \Rightarrow y = 6 \times -\tfrac{1}{4}(x-2)^{-4} + c = -\tfrac{3}{2}(x-2)^{-4} + c$$

When $x = 3$, $y = \frac{5}{2}$:

$$\tfrac{5}{2} = -\tfrac{3}{2}(3-2)^{-4} + c \Rightarrow \tfrac{5}{2} = -\tfrac{3}{2} + c \Rightarrow c = 4$$

Therefore $y = -\frac{3}{2}(x-2)^{-4} + 4$.

4)

$$\frac{dy}{dx} = \sqrt[3]{4+x} \Rightarrow \frac{dy}{dx} = (4+x)^{1/3} \Rightarrow y = \tfrac{3}{4}(4+x)^{4/3} + c$$

When $x = 4$, $y = 3$:

$$3 = \tfrac{3}{4}(4+4)^{4/3} + c \Rightarrow 3 = 12 + c \Rightarrow c = -9$$

Therefore $y = \frac{3}{4}(4+x)^{4/3} - 9$.

5 a)

$$\int_2^3 (x-2)^8\,dx = [\tfrac{1}{9}(x-2)^9]_2^3 = (\tfrac{1}{9}(3-2)^9) - (\tfrac{1}{9}(2-2)^9)$$
$$= (\tfrac{1}{9}\times 1) - (\tfrac{1}{9}\times 0) = \tfrac{1}{9}$$

5 b)

$$\int_{-6}^{-3} 24(6+x)^5\,dx = [24\times\tfrac{1}{6}(6+x)^6]_{-6}^{-3} = [4(6+x)^6]_{-6}^{-3}$$
$$= (4(6-3)^6) - (4(6-6)^6) = (4\times 729) - (4\times 0) = 2916$$

5 c)

$$\int_{-1}^{1} \sqrt{x+3}\,dx = \int_{-1}^{1} (x+3)^{1/2}\,dx = [\tfrac{2}{3}(x+3)^{3/2}]_{-1}^{1}$$
$$= (\tfrac{2}{3}(1+3)^{3/2}) - (\tfrac{2}{3}(-1+3)^{3/2})$$
$$= (\tfrac{2}{3}\times 8) - (\tfrac{2}{3}\times 2\sqrt{2}) = \tfrac{16}{3} - \tfrac{4\sqrt{2}}{3}$$

5 d)

$$\int_{-3}^{4} \frac{44}{\sqrt[3]{4+x}}\,dx = \int_{-3}^{4} 44(4+x)^{-1/3}\,dx$$
$$= [44\times\tfrac{3}{2}(4+x)^{2/3}]_{-3}^{4} = [66(4+x)^{2/3}]_{-3}^{4}$$
$$= (66(4+4)^{2/3}) - (66(4-3)^{2/3})$$
$$= (66\times 4) - (66\times 1) = (264) - (66) = 198$$

6 a)

$$\lim_{\delta x\to 0} \sum_{x=-5}^{26} (6+x)^{3/5}\delta x = \int_{-5}^{26} (6+x)^{3/5}\,dx = [\tfrac{5}{8}(6+x)^{8/5}]_{-5}^{26}$$
$$= (\tfrac{5}{8}(6+26)^{8/5}) - (\tfrac{5}{8}(6-5)^{8/5})$$
$$= (\tfrac{5}{8}\times 256) - (\tfrac{5}{8}\times 1) = (160) - (\tfrac{5}{8}) = \tfrac{1275}{8}$$

6 b)

$$\lim_{\delta x\to 0} \sum_{x=19}^{84} \frac{4}{\sqrt[4]{x-3}}\delta x = \int_{19}^{84} 4(x-3)^{-1/4}\,dx$$
$$= [4\times\tfrac{4}{3}(x-3)^{3/4}]_{19}^{84} = [\tfrac{16}{3}(x-3)^{3/4}]_{19}^{84}$$
$$= (\tfrac{16}{3}(84-3)^{3/4}) - (\tfrac{16}{3}(19-3)^{3/4})$$
$$= (\tfrac{16}{3}\times 27) - (\tfrac{16}{3}\times 8) = (144) - (\tfrac{128}{3}) = \tfrac{304}{3}$$

7 a)

$\int (2x+1)^4\,dx = \frac{1}{2} \times \frac{1}{5}(2x+1)^5 + c = \frac{1}{10}(2x+1)^5 + c$

7 b)

$\int (2-3x)^5\,dx = -\frac{1}{3} \times \frac{1}{6}(2-3x)^6 + c = -\frac{1}{18}(2-3x)^6 + c$

7 c)

$\int 4(5+6x)^9\,dx = 4 \times \frac{1}{6} \times \frac{1}{10}(5+6x)^{10} + c = \frac{1}{15}(5+6x)^{10} + c$

7 d)

$$\begin{aligned}\int (3(2x-3)^3 - 4(4-5x)^4)\,dx &= 3 \times \tfrac{1}{2} \times \tfrac{1}{4}(2x-3)^4 - 4 \times -\tfrac{1}{5} \times \tfrac{1}{5}(4-5x)^5 + c \\ &= \tfrac{3}{8}(2x-3)^4 + \tfrac{4}{25}(4-5x)^5 + c\end{aligned}$$

8 a)

$\frac{dy}{dx} = (5x+9)^{7/3} \Rightarrow y = \frac{1}{5} \times \frac{3}{10}(5x+9)^{10/3} + c \Rightarrow y = \frac{3}{50}(5x+9)^{10/3} + c$

8 b)

$\frac{dy}{dx} = (6-x)^{-5} \Rightarrow y = -1 \times -\frac{1}{4}(6-x)^{-4} + c \Rightarrow y = \frac{1}{4}(6-x)^{-4} + c$

8 c)

$\frac{dy}{dx} = \sqrt{10+3x} \Rightarrow \frac{dy}{dx} = (10+3x)^{1/2} \Rightarrow y = \frac{1}{3} \times \frac{2}{3}(10+3x)^{3/2} + c \Rightarrow y = \frac{2}{9}(10+3x)^{3/2} + c$

8 d)

$$\frac{dy}{dx} = \frac{20}{\sqrt[3]{1-x}} \Rightarrow \frac{dy}{dx} = 20(1-x)^{-1/3} \Rightarrow y = 20 \times -1 \times \tfrac{3}{2}(1-x)^{2/3} + c \Rightarrow y = -30(1-x)^{2/3} + c$$

9)

$$\frac{dy}{dx} = \frac{6}{(3x-2)^2} \Rightarrow \frac{dy}{dx} = 6(3x-2)^{-2} \Rightarrow y = 6 \times \tfrac{1}{3} \times -1 \times (3x-2)^{-1} + c = -2(3x-2)^{-1} + c$$

When $x = 0$, $y = 2$:

$$2 = -2(3 \times 0 - 2)^{-1} + c \Rightarrow 2 = 1 + c \Rightarrow c = 1$$

$$y = -2(3x-2)^{-1} + 1 = -\frac{2}{3x-2} + 1 = \frac{3x-2}{3x-2} - \frac{2}{3x-2} \Rightarrow y = \frac{3x-4}{3x-2}$$

10)

$$\frac{dy}{dx} = \frac{14}{(4x+1)^{9/2}} \Rightarrow \frac{dy}{dx} = 14(4x+1)^{-9/2} \Rightarrow y = 14 \times \tfrac{1}{4} \times -\tfrac{2}{7}(4x+1)^{-7/2} + c = -(4x+1)^{-7/2} + c$$

When $x = 0$, $y = 2$:

$$2 = -(4 \times 0 + 1)^{-7/2} + c \Rightarrow 2 = -1 + c \Rightarrow c = 3$$

Therefore $y = -(4x+1)^{-7/2} + 3 \Rightarrow y = 3 - \frac{1}{(4x+1)^{7/2}}$.

11 a)

$$\begin{aligned}\int_2^3 (2x-5)^5\,dx &= [\tfrac{1}{2}\times\tfrac{1}{6}(2x-5)^6]_2^3 = [\tfrac{1}{12}(2x-5)^6]_2^3\\ &= (\tfrac{1}{12}(2\times 3-5)^6) - (\tfrac{1}{12}(2\times 2-5)^6)\\ &= (\tfrac{1}{12}\times 1) - (\tfrac{1}{12}\times 1) = 0\end{aligned}$$

11 b)

$$\begin{aligned}\int_{-1/4}^{1/8} \sqrt{3+8x}\,dx &= \int_{-1/4}^{1/8} (3+8x)^{1/2}\,dx\\ &= [\tfrac{1}{8}\times\tfrac{2}{3}(3+8x)^{3/2}]_{-1/4}^{1/8} = [\tfrac{1}{12}(3+8x)^{3/2}]_{-1/4}^{1/8}\\ &= (\tfrac{1}{12}(3+8\times\tfrac{1}{8})^{3/2}) - (\tfrac{1}{12}(3+8\times(-\tfrac{1}{4}))^{3/2})\\ &= (\tfrac{1}{12}\times 8) - (\tfrac{1}{12}\times 1) = (\tfrac{2}{3}) - (\tfrac{1}{12}) = \tfrac{7}{12}\end{aligned}$$

11 c)

$$\begin{aligned}\int_1^2 \frac{15}{(5-6x)^2}\,dx &= \int_1^2 15(5-6x)^{-2}\,dx\\ &= [15\times -\tfrac{1}{6}\times -1\times (5-6x)^{-1}]_1^2 = [\tfrac{5}{2}(5-6x)^{-1}]_1^2\\ &= (\tfrac{5}{2}(5-6\times 2)^{-1}) - (\tfrac{5}{2}(5-6\times 1)^{-1})\\ &= (\tfrac{5}{2}\times -\tfrac{1}{7}) - (\tfrac{5}{2}\times -1) = (-\tfrac{5}{14}) - (-\tfrac{5}{2}) = \tfrac{15}{7}\end{aligned}$$

11 d)

$$\begin{aligned}\int_{5/16}^{3/2} \frac{8}{\sqrt[3]{3+16x}}\,dx &= \int_{5/16}^{3/2} 8(3+16x)^{-1/3}\,dx\\ &= [8\times\tfrac{1}{16}\times\tfrac{3}{2}(3+16x)^{2/3}]_{5/16}^{3/2} = [\tfrac{3}{4}(3+16x)^{2/3}]_{5/16}^{3/2}\\ &= (\tfrac{3}{4}(3+16\times\tfrac{3}{2})^{2/3}) - (\tfrac{3}{4}(3+16\times\tfrac{5}{16})^{2/3})\\ &= (\tfrac{3}{4}\times 9) - (\tfrac{3}{4}\times 4) = (\tfrac{27}{4}) - (3) = \tfrac{15}{4}\end{aligned}$$

12 a)

$$\begin{aligned}\lim_{\delta x\to 0}\sum_{x=-1/3}^{10} \tfrac{2}{5}(2+3x)^{3/5}\delta x &= \int_{-1/3}^{10} \tfrac{2}{5}(2+3x)^{3/5}\,dx\\ &= [\tfrac{2}{5}\times\tfrac{1}{3}\times\tfrac{5}{8}(2+3x)^{8/5}]_{-1/3}^{10} = [\tfrac{1}{12}(2+3x)^{8/5}]_{-1/3}^{10}\\ &= (\tfrac{1}{12}(2+3\times 10)^{8/5}) - (\tfrac{1}{12}(2+3\times(-\tfrac{1}{3}))^{8/5})\\ &= (\tfrac{1}{12}\times 256) - (\tfrac{1}{12}\times 1) = (\tfrac{64}{3}) - (\tfrac{1}{12}) = \tfrac{85}{4}\end{aligned}$$

12 b)

$$\lim_{\delta x\to 0}\sum_{x=1/9}^{1/3}\frac{243}{(4-9x)^4}\delta x=\int_{1/9}^{1/3}243(4-9x)^{-4}\,dx$$
$$=[243\times-\tfrac{1}{9}\times-\tfrac{1}{3}(4-9x)^{-3}]_{1/9}^{1/3}=[9(4-9x)^{-3}]_{1/9}^{1/3}$$
$$=(9(4-9\times\tfrac{1}{3})^{-3})-(9(4-9\times\tfrac{1}{9})^{-3})$$
$$=(9\times 1)-(9\times\tfrac{1}{27})=(9)-(\tfrac{1}{3})=\tfrac{26}{3}$$

13 a)

$\int 2x(x^2+3)^5\,dx=2x\times\frac{1}{2x}\times\frac{1}{6}(x^2+3)^6+c=\frac{1}{6}(x^2+3)^6+c$

13 b)

$\int 12x^2(4x^3-5)^3\,dx=12x^2\times\frac{1}{12x^2}\times\frac{1}{4}(4x^3-5)^4+c=\frac{1}{4}(4x^3-5)^4+c$

13 c)

$\int x(2x^2-5)^4\,dx=x\times\frac{1}{4x}\times\frac{1}{5}(2x^2-5)^5+c=\frac{1}{20}(2x^2-5)^5+c$

13 d)

$$\int 2x^3\sqrt{x^4-3}\,dx=\int 2x^3(x^4-3)^{1/2}\,dx=2x^3\times\frac{1}{4x^3}\times\frac{2}{3}(x^4-3)^{3/2}+c=\frac{1}{3}(x^4-3)^{3/2}+c$$

14 a)

$$\frac{dy}{dx}=9x^2(3x^3+2)^{-2}\Rightarrow y=9x^2\times\frac{1}{9x^2}\times-1\times(3x^3+2)^{-1}+c\Rightarrow y=-(3x^3+2)^{-1}+c$$

14 b)

$$\frac{dy}{dx}=10x^3(2-5x^4)^{5/2}\Rightarrow y=10x^3\times-\frac{1}{20x^3}\times\frac{2}{7}(2-5x^4)^{7/2}+c\Rightarrow y=\frac{1}{7}(2-5x^4)^{7/2}+c$$

14 c)

$$\frac{dy}{dx}=x\sqrt[3]{3+4x^2}\Rightarrow\frac{dy}{dx}=x(3+4x^2)^{1/3}\Rightarrow y=x\times\frac{1}{8x}\times\frac{3}{4}(3+4x^2)^{4/3}+c\Rightarrow y=\frac{3}{32}(3+4x^2)^{4/3}+c$$

14 d)

$$\frac{dy}{dx}=\frac{100x^8}{(2x^9+3)^6}\Rightarrow\frac{dy}{dx}=100x^8(2x^9+3)^{-6}$$
$$\Rightarrow y=100x^8\times\frac{1}{18x^8}\times-\frac{1}{5}(2x^9+3)^{-5}+c$$
$$\Rightarrow y=-\frac{10}{9}(2x^9+3)^{-5}+c$$

15)

$$\frac{dy}{dx} = 5x(5x^2-1)^7 \Rightarrow y = 5x \times \frac{1}{10x} \times \tfrac{1}{8}(5x^2-1)^8 + c \Rightarrow y = \tfrac{1}{16}(5x^2-1)^8 + c$$

When $x = 0$, $y = 1$:

$$1 = \tfrac{1}{16} \times (5 \times 0^2 - 1)^8 + c \Rightarrow 1 = \tfrac{1}{16} + c \Rightarrow c = \tfrac{15}{16}$$

Therefore $y = \frac{1}{16}(5x^2-1)^8 + \frac{15}{16}$.

16)

$$\frac{dy}{dx} = \frac{9x^2}{\sqrt{27x^3-1}} \Rightarrow \frac{dy}{dx} = 9x^2(27x^2-1)^{-1/2}$$

$$y = 9x^2 \times \frac{1}{81x^2} \times 2(27x^3-1)^{1/2} + c \Rightarrow y = \tfrac{2}{9}(27x^3-1)^{1/2} + c$$

When $x = \frac{1}{3}$, $y = 0$:

$$0 = \tfrac{2}{9} \times (27 \times (\tfrac{1}{3})^3 - 1)^{1/2} + c \Rightarrow 0 = 0 + c \Rightarrow c = 0$$

Therefore $y = \frac{2}{9}(27x^3-1)^{1/2}$.

17 a)

$$\begin{aligned}\int_0^2 x(10x^2+1)^4\,dx &= [x \times \tfrac{1}{20x} \times \tfrac{1}{5}(10x^2+1)^5]_0^2 = [\tfrac{1}{100}(10x^2+1)^5]_0^2 \\ &= (\tfrac{1}{100} \times (10 \times 2^2 + 1)^5) - (\tfrac{1}{100} \times (10 \times 0^2 + 1)^5) \\ &= (\tfrac{1}{100} \times 115856201) - (\tfrac{1}{100} \times 1) = \tfrac{115856201}{100} - \tfrac{1}{100} = 1158562\end{aligned}$$

17 b)

$$\begin{aligned}\int_{1/\sqrt{2}}^{3/2} x(4x^2-1)^{2/3}\,dx &= [x \times \tfrac{1}{8x} \times \tfrac{3}{5}(4x^2-1)^{5/3}]_{1/\sqrt{2}}^{3/2} \\ &= [\tfrac{3}{40}(4x^2-1)^{5/3}]_{1/\sqrt{2}}^{3/2} \\ &= (\tfrac{3}{40} \times (4 \times (\tfrac{3}{2})^2 - 1)^{5/3}) - (\tfrac{3}{40} \times (4 \times (\tfrac{1}{\sqrt{2}})^2 - 1)^{5/3}) \\ &= (\tfrac{3}{40} \times 32) - (\tfrac{3}{40} \times 1) = (\tfrac{12}{5}) - (\tfrac{3}{40}) = \tfrac{93}{40}\end{aligned}$$

17 c)

$$\begin{aligned}\int_2^3 \frac{6x^2}{(x^3+2)^2}\,dx &= \int_2^3 6x^2(x^3+2)^{-2}\,dx \\ &= [6x^2 \times \tfrac{1}{3x^2} \times -1 \times (x^3+2)^{-1}]_2^3 = [-2(x^3+2)^{-1}]_2^3 \\ &= (-2 \times (3^3+2)^{-1}) - (-2 \times (2^3+2)^{-1}) \\ &= (-2 \times \tfrac{1}{29}) - (-2 \times \tfrac{1}{10}) = (-\tfrac{2}{29}) - (-\tfrac{1}{5}) = \tfrac{19}{145}\end{aligned}$$

17 d)

$$\begin{aligned}\int_0^{1/2} \frac{800x^3}{(1-x^4)^3}\,dx &= \int_0^{1/2} 800x^3(1-x^4)^{-3}\,dx \\ &= [800x^3 \times -\tfrac{1}{4x^3} \times -\tfrac{1}{2}(1-x^4)^{-2}]_0^{1/2} = [100(1-x^4)^{-2}]_0^{1/2} \\ &= (100 \times (1-(\tfrac{1}{2})^4)^{-2}) - (100 \times (1-0^4)^{-2}) \\ &= (100 \times \tfrac{256}{225}) - (100 \times 1) = (\tfrac{1024}{9}) - (100) = \tfrac{124}{9}\end{aligned}$$

18 a)

$$\begin{aligned}\lim_{\delta x \to 0} \sum_{x=0}^{1} 98x(1+14x^2)^{5/2}\delta x &= \int_0^1 98x(1+14x^2)^{5/2}\,dx \\ &= [98x \times \tfrac{1}{28x} \times \tfrac{2}{7}(1+14x^2)^{7/2}]_0^1 = [(1+14x^2)^{7/2}]_0^1 \\ &= ((1+14\times 1^2)^{7/2}) - ((1+14\times 0^2)^{7/2}) \\ &= (15^{7/2}) - (1) = 3375\sqrt{15} - 1\end{aligned}$$

18 b)

$$\begin{aligned}\lim_{\delta x \to 0} \sum_{x=0}^{9} \frac{\sqrt{\sqrt{x}+1}}{\sqrt{x}}\delta x &= \int_0^9 x^{-1/2}(x^{1/2}+1)^{1/2}\,dx \\ &= [x^{-1/2} \times \tfrac{1}{\frac{1}{2}x^{-1/2}} \times \tfrac{2}{3}(x^{1/2}+1)^{3/2}]_0^9 = [\tfrac{4}{3}(x^{1/2}+1)^{3/2}]_0^9 \\ &= (\tfrac{4}{3} \times (9^{1/2}+1)^{3/2}) - (\tfrac{4}{3} \times (0^{1/2}+1)^{3/2}) \\ &= (\tfrac{4}{3} \times 8) - (\tfrac{4}{3} \times 1) = (\tfrac{32}{3}) - (\tfrac{4}{3}) = \tfrac{28}{3}\end{aligned}$$

19)

$$\begin{aligned}\int_{-1}^{0} \tfrac{1}{2}x(x^2-1)^5\,dx &= [\tfrac{1}{2}x \times \tfrac{1}{2x} \times \tfrac{1}{6}(x^2-1)^6]_{-1}^0 = [\tfrac{1}{24}(x^2-1)^6]_{-1}^0 \\ &= (\tfrac{1}{24} \times (0^2-1)^6) - (\tfrac{1}{24} \times ((-1)^2-1)^6) \\ &= (\tfrac{1}{24} \times 1) - (\tfrac{1}{24} \times 0) = (\tfrac{1}{24}) - (0) = \tfrac{1}{24}\end{aligned}$$

$$\begin{aligned}\int_0^{1} \tfrac{1}{2}x(x^2-1)^5\,dx &= [\tfrac{1}{24}(x^2-1)^6]_0^1 \\ &= (\tfrac{1}{24} \times (1^2-1)^6) - (\tfrac{1}{24} \times (0^2-1)^6) \\ &= (\tfrac{1}{24} \times 0) - (\tfrac{1}{24} \times 1) = (0) - (\tfrac{1}{24}) = -\tfrac{1}{24}\end{aligned}$$

Therefore Area $= \frac{1}{24} - (-\frac{1}{24}) = \frac{1}{12}$.

20)

$$\begin{aligned}
&\int_0^{1/2} \frac{2}{(x+1)^2}\,dx - \int_0^{1/2} \frac{2}{(2x+1)^2}\,dx \\
&\qquad = \int_0^{1/2} (\tfrac{2}{(x+1)^2} - \tfrac{2}{(2x+1)^2})\,dx \\
&\qquad = \int_0^{1/2} (2(x+1)^{-2} - 2(2x+1)^{-2})\,dx \\
&\qquad = [2 \times -1 \times (x+1)^{-1} - 2 \times \tfrac{1}{2} \times -1 \times (2x+1)^{-1}]_0^{1/2} \\
&\qquad = [-2(x+1)^{-1} + (2x+1)^{-1}]_0^{1/2} \\
&\qquad = (-2 \times (\tfrac{1}{2}+1)^{-1} + (2 \times \tfrac{1}{2}+1)^{-1}) \\
&\qquad\quad - (-2 \times (0+1)^{-1} + (2 \times 0+1)^{-1}) \\
&\qquad = (-2 \times \tfrac{2}{3} + \tfrac{1}{2}) - (-2 \times 1 + 1) = (-\tfrac{5}{6}) - (-1) = \tfrac{1}{6}
\end{aligned}$$

21)

$$\frac{12x - 9x^2}{(2x^2 - x^3 + 1)^3} = 0 \Rightarrow 12x - 9x^2 = 0 \Rightarrow 4x - 3x^2 = 0 \Rightarrow x(4 - 3x) = 0 \Rightarrow x = 0 \text{ or } x = \tfrac{4}{3}$$

$$\begin{aligned}
\int_0^{4/3} \frac{12x - 9x^2}{(2x^2 - x^3 + 1)^3}\,dx &= \int_0^{4/3} (12x - 9x^2)(2x^2 - x^3 + 1)^{-3}\,dx \\
&= [(12x - 9x^2) \times \tfrac{1}{4x - 3x^2} \times -\tfrac{1}{2}(2x^2 - x^3 + 1)^{-2}]_0^{4/3} \\
&= [-\tfrac{12x - 9x^2}{8x - 6x^2}(2x^2 - x^3 + 1)^{-2}]_0^{4/3} \\
&= [-\tfrac{3(4x - 3x^2)}{2(4x - 3x^2)}(2x^2 - x^3 + 1)^{-2}]_0^{4/3} \\
&= [-\tfrac{3}{2}(2x^2 - x^3 + 1)^{-2}]_0^{4/3} \\
&= (-\tfrac{3}{2} \times (2 \times (\tfrac{4}{3})^2 - (\tfrac{4}{3})^3 + 1)^{-2}) \\
&\quad - (-\tfrac{3}{2} \times (2 \times 0^2 - 0^3 + 1)^{-2}) \\
&= (-\tfrac{3}{2} \times \tfrac{729}{3481}) - (-\tfrac{3}{2} \times 1) \\
&= (-\tfrac{2187}{6962}) - (-\tfrac{3}{2}) = \tfrac{6315}{3481}
\end{aligned}$$